

Z-basis metrics for parameterized quantum circuits: exact identities, structural blind spots, pole-damped optimization, and a relaxation-aware entropy benchmark

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Abstract

The Qang (qg) framework [1] expresses single-qubit rotations through two measurement-level quantities, the polar bias $qg_Z = \langle \sigma_z \rangle$ and the outcome entropy $qg_S = H(p_0)$. We report results obtained with the open-source reference implementation **qang**. First, the two metrics are linked by an exact, branch-free identity, $qg_S = H((1 + qg_Z)/2)$, which holds for any single-qubit state. Consequently, any new multi-qubit information must come from the joint outcome distribution. We name the resulting Z-basis total correlation and give its bounds and its scope: it is not an entanglement measure. Second, we identify exact blind spots: graph-state entanglement and readout-only dephasing are invisible to every Z-diagonal statistic, and qg-space updates cannot reach optima outside $[0, \pi]$ because of the arccos range. On H_2 they recover none of the correlation energy. Third, we study a pole-damped gradient step whose damping schedule is $|dqg_Z/d\theta|$. On a 300-landscape benchmark, H_2 , and a 4-qubit LiH Hamiltonian with a mixed R_y/R_x ansatz, it trades $3\text{--}20\times$ more iterations at well-tuned learning rates for convergence where plain gradient descent diverges. Fourth, as a benchmark, qg_S tracks heavy output probability ($r = -0.93$) and needs no calibration, unlike linear XEB. However, it is *not* monotonic under amplitude damping: under device-calibrated noise with added idle relaxation it falls below its own noiseless value. Pairing qg_S with the register-averaged qg_Z removes the ambiguity in 23 of 24 circuits tested. A summary section lists twenty-five follow-up studies (QML encoding, control quantization, noise-type detection, circuit knitting, chemistry, error mitigation, thermal states, error correction, Hubbard dynamics, constrained QAOA, qubit characterization, syndrome tracking, coherence, few-shot estimation and IonQ noise models), each with its classical or standard control; the symmetry-witness results are developed in a companion paper. Every number is reproduced by a script and pinned by a regression test in the public repository.

1 Introduction

Parameterized quantum circuits are written in angles, but everything a device returns is a distribution over measurement outcomes. The Qang framework [1] makes that second view the primary one: for a qubit at polar angle θ ,

$$qg_Z(\theta) = \cos \theta = \langle \sigma_z \rangle \in [-1, 1], \quad qg_S(\theta) = H(\cos^2 \frac{\theta}{2}) \in [0, 1], \quad (1)$$

with $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$. This paper collects results obtained with the reference implementation **qang**, with the aim of stating where the metrics are exact, where they are blind, and where they mislead. Section 2 gives exact identities, Section 3 the structural blind spots, Section 4 a pole-damped optimizer on molecular Hamiltonians, and Section 5 the behavior of

qg_S as a benchmark, including a device-calibrated validation. Section 6 summarizes later results, and Section 7 lists the limitations together.

All results are produced by scripts in the `examples/` folder of the repository and pinned by regression tests (Section 8). Molecular Hamiltonians were derived once with PySCF [2] and are hard-coded, so neither PySCF nor qiskit-nature is a dependency.

2 Exact identities

2.1 A branch-free link between qg_Z and qg_S

Inverting $qg_S \mapsto \theta$ requires choosing a half-domain, because $qg_S(\theta) = qg_S(\pi - \theta)$. The opposite direction needs no choice.

Proposition 1. *For any single-qubit density matrix ρ , with $qg_Z = \text{Tr}(\rho\sigma_z)$ and $qg_S = H(\langle 0|\rho|0\rangle)$,*

$$qg_S = H\left(\frac{1 + qg_Z}{2}\right), \quad \frac{d qg_S}{d qg_Z} = \frac{1}{2} \log_2 \frac{1 - qg_Z}{1 + qg_Z}. \quad (2)$$

Proof. $qg_Z = p_0 - p_1 = 2p_0 - 1$ with $p_0 = \langle 0|\rho|0\rangle$; substitute into $H(p_0)$ and differentiate. $\square$

The proof uses only the Z-diagonal population. The identity therefore holds however the state was prepared (R_y , R_x , a noisy channel, or a partial trace of a larger register), and the tests check it on random Haar states. The derivative diverges at $qg_Z = \pm 1$. This is the entropy-side counterpart of the Jacobian singularity $d\theta/dqg_Z = -1/\sin\theta$.

2.2 Z-basis total correlation

Applied to each reduced single-qubit state, Proposition 1 makes the per-qubit qg_Z profile and the per-qubit marginal qg_S two encodings of the same information. Anything new about an n -qubit register must come from the joint distribution of the Z-basis outcomes $X_1, \dots, X_n$. We define

$$C_Z(\rho) = \sum_{i=1}^n H(X_i) - H(X_1, \dots, X_n), \quad (3)$$

Watanabe's total correlation [3] of the outcomes. For $n = 2$ it equals the mutual information. By subadditivity, $0 \leq C_Z \leq n - 1$ bits, with $C_Z = 0$ iff the outcomes are independent. Product states give 0, Bell states 1 and GHZ $_n$ states $n - 1$ (the maximum).

Remark 1 (Scope). C_Z is a classical correlation measure and is not a witness of entanglement in either direction. The separable mixture $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$ has $C_Z = 1$, as does a Bell state. Every graph state has $C_Z = 0$ (Section 3).

3 Structural blind spots

Graph states. A graph state [4] is $H^{\otimes n}$ followed by CZ gates. $H^{\otimes n}$ makes every $|\text{amplitude}|^2$ equal to 2^{-n} , and CZ is diagonal. For every graph, therefore, each qubit has $qg_Z = 0$, the joint qg_S is maximal, and $C_Z = 0$. The entanglement, certified independently through the stabilizers $X_i \prod_{j \in N(i)} Z_j$, lives entirely in phases.

Readout dephasing. Dephasing preserves the diagonal of ρ . Applied only before measurement, it therefore leaves qg_S exactly unchanged at every strength, including full dephasing (checked to machine precision on a 4-qubit Quantum Volume circuit). The same channel applied *inside* the circuit is visible: later entangling gates turn the lost coherence into randomized populations, and qg_S rises monotonically.

Table 1: Success rates over 300 random one-qubit landscapes per learning rate, started near a pole.

η	0.3	1.0	2.0	3.0	5.0	10.0
plain success	0.997	0.773	0.187	0.087	0.057	0.027
damped success	0.963	1.000	0.477	0.357	0.253	0.130
damped trapped	0.017	0.000	0.030	0.023	0.040	0.067

The arccos range. A qg-space parameter update is mapped back through $\theta = \arccos(qg_Z) \in [0, \pi]$. On the H_2 Hamiltonian (0.735 Å, STO-3G, two qubits after parity reduction), the single-excitation optimum lies at $\theta \approx -0.22$. Starting from Hartree–Fock, raw, clipped and Tikhonov-regularized qg-space descent all stop at exactly the Hartree–Fock energy (-1.1169990 Ha, error 2.0×10^{-2} Ha). They recover none of the correlation energy, whereas θ -space descent reaches the FCI energy to 10^{-9} Ha. Regularizing the Jacobian cannot fix this, because the limit comes from the range of arccos, not from its derivative.

4 Pole-damped gradient descent

To keep updates in θ -space while still using the pole geometry, we damp each parameter’s step by $|dqg_Z/d\theta_i|$, floored at ε :

$$\theta_i \leftarrow \theta_i - \eta \max(|\sin \theta_i|, \varepsilon) \frac{\partial E}{\partial \theta_i}. \quad (4)$$

This is position-dependent trust-region damping in the Levenberg–Marquardt tradition [5, 6]. It is not a new class of optimizer; the only framework-specific element is that the schedule is derived from the geometry rather than tuned. Because the factor depends only on populations, and $R_x(\theta)|0\rangle$ and $R_y(\theta)|0\rangle$ have identical populations, it applies unchanged to either axis. We use $\varepsilon = 0.05$ throughout; results are insensitive to this choice.

Random landscapes. For 300 random landscapes $E = h_z \cos \theta + h_x \sin \theta$ per learning rate, started near a pole (Table 1), damping costs a few percent of success at a safe rate. At $\eta = 1$ it succeeds in 100% of cases versus 77%, and at larger rates it succeeds 2.5–5 times more often. The cost is a trapping mode, in which the optimizer oscillates at its starting pole; it reaches 6.7% at $\eta = 10$.

H_2 , two coupled parameters. Ansatz $R_y(\theta_0) \otimes R_y(\theta_1)$ followed by CX; the optimum has $\theta_0 = \pi$ (a pole). From $(1.0, 10^{-6})$, damping needs about $3\times$ more steps to reach chemical accuracy at safe rates (603 vs. 196 at $\eta = 0.05$). At $\eta = 2.5, 3$ and 4, plain descent never reaches chemical accuracy, while the damped version reaches the FCI energy in 10, 8 and 6 steps.

LiH, mixed axes. LiH at 1.5459 Å (STO-3G, 2 electrons in 3 spatial orbitals, parity mapping) gives 4 qubits and 52 Pauli terms, the scale of early hardware VQE work [7, 8]. The ansatz is $R_y(\theta_0)$, $R_y(\theta_1)$ on the occupied orbitals and $R_y(\theta_2)$, $R_x(\theta_3)$ on the virtual ones, followed by CX($2 \rightarrow 0$) and CX($3 \rightarrow 1$). At $(\pi, \pi, 0, 0)$ its state has overlap 1 with qiskit-nature’s Hartree–Fock circuit. All four optimal parameters lie within 0.04 rad of a pole, which makes the cost larger than for H_2 : 355 vs. 17 steps at $\eta = 0.3$. The benefit is also larger (Table 2). The R_x phase is physical: after the CX it changes the energy relative to R_y at the same angle, although the single-qubit populations are identical.

Table 2: LiH, start $(\pi, \pi, 10^{-6}, 10^{-6})$: steps to sustained convergence (10^{-5} Ha) to the ansatz optimum -0.0440152 Ha (active-space electronic energy; HF -0.0437813 , FCI -0.0448309).

η	0.3	1.0	2.0	2.5	4.0	7.0	10.0
plain θ	17	5	never	never	never	never	never
pole-damped	355	106	53	42	26	15	10

5 qg_S as a benchmark

Heavy output probability and XEB. On 10 random 4-qubit Quantum Volume circuits [9] mixed with global depolarizing noise, qg_S and heavy output probability have Pearson $r = -0.926$ over 110 points. Linear XEB [10] has a noiseless value $A - 1$, with $A = 2^n \sum_x p(x)^2$ ranging from 1.23 to 2.33 on the same circuits family ($n = 3-6$), so it needs circuit-specific calibration; qg_S does not. Conversely, XEB’s estimator is a sample mean and is unbiased at any shot count, while the plug-in entropy is biased low (bias -0.054 at 50 shots on 4 qubits). The Miller-Madow correction [11] reduces that bias about fivefold.

Amplitude damping breaks monotonicity. The reading “more noise $\Rightarrow$ higher qg_S ” holds only for channels whose fixed point is maximally mixed. With amplitude damping applied after every two-qubit block (Fig. 1a), qg_S rises from 0.919 to 0.983 at $\gamma = 0.1$ and then falls to 0 at $\gamma = 1$. At $\gamma = 0.4$ it reads 0.918, almost the noiseless value.

The fix: register-averaged qg_Z . The mean $q\bar{g}_Z = \frac{1}{n} \sum_i \text{Tr}(\rho_i \sigma_z)$ is pulled towards 0 by unital noise and towards +1 by relaxation. At $\gamma = 0$ and 0.4 the pairs $(qg_S, q\bar{g}_Z)$ are (0.919, -0.029) and (0.918, $+0.310$). $q\bar{g}_Z$ rose monotonically along a 21-point γ grid in 23 of 24 circuits ($n = 3, 4, 5$; seeds 0–7); the exception dips by 0.015 before rising. $q\bar{g}_Z$ is linear in ρ , so its estimator from counts is an exact sample mean (Appendix A).

Device-calibrated validation. We ran the same circuit through Qiskit Runtime on the `fake_brisbane` backend, a simulator whose noise model is built from a real IBM device’s calibration data. An idle delay before readout served as a T1 knob (Fig. 1b). Heavy output probability fell monotonically (0.685 to 0.448) and $q\bar{g}_Z$ rose monotonically (-0.036 to $+0.370$). qg_S rose to 0.977 at 50 μs and then fell to 0.892 at 200 μs , below its noiseless value of 0.919. Read on its own, qg_S would rank the most relaxed operating point as the least noisy.

The same script evaluated the LiH ansatz without error mitigation. The raw error was $+0.036$ Ha at Hartree-Fock and $+0.035$ Ha at the optimum. This is about $20\times$ chemical accuracy and about $150\times$ the 0.23 mHa difference the optimizer resolves noiselessly. A hardware run of this VQE therefore needs mitigation first.

Randomized benchmarking and ZNE. Under per-gate depolarizing noise p , the RB survival signal is exactly $qg_Z(m) = (1 - p)^{m+1}$, identical across random Clifford sequences to 10 digits in our runs [12]. For zero-noise extrapolation [13], extrapolation in qg_Z -space is exact for Bloch-vector shrinkage but biased (1.8×10^{-3}) for coherent angle drift; θ -space shows the reverse. One should extrapolate in whichever space makes the noise linear. In barren-plateau settings [14, 15], regularized qg -space gradients keep the exponential variance decay, and raw qg -space gradients still diverge at the poles.

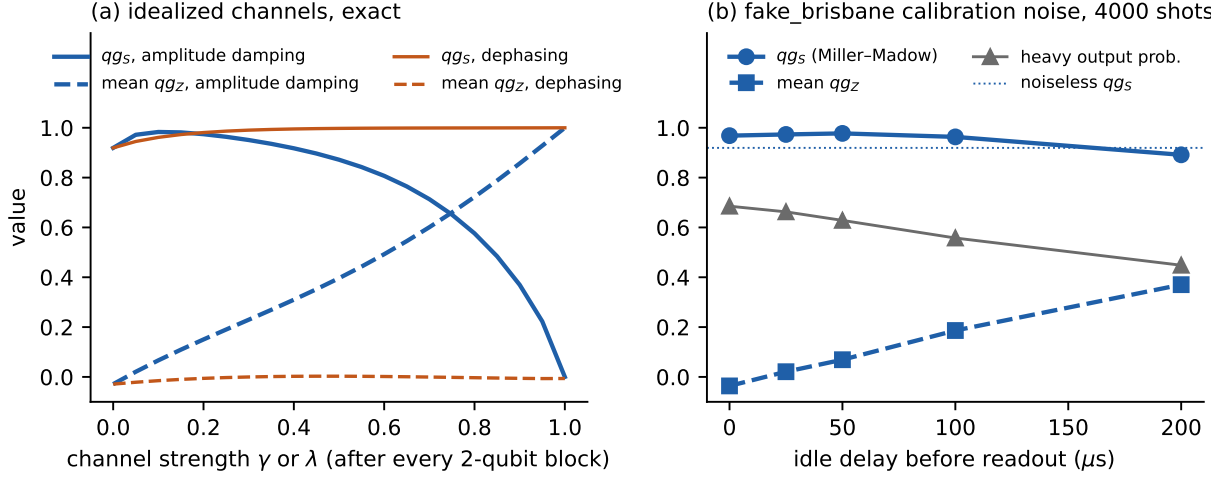


Figure 1: (a) 4-qubit Quantum Volume circuit, exact density-matrix simulation with amplitude damping or dephasing after every two-qubit block. qgs is non-monotonic under amplitude damping, while the mean qgz separates the regimes. (b) The same circuit under **fake_brisbane** calibration noise with an idle delay before readout. Heavy output probability and mean qgz are monotonic; qgs falls below its noiseless value (dotted).

6 Further results (summary)

After the first draft, the same code base was used for twenty-five further studies. Each compares a qg-based choice with the standard one and with a classical control, and each is documented in `RESEARCH_NOTES.md` (section numbers below) with a script and pinned tests. Table 3 lists them. The column “gain” states against what the qg choice wins, if at all. None of them is a quantum advantage over classical computation. Where a classical method was run, it matched or beat the quantum model at these sizes. The symmetry-witness studies (§20, §21, §24, §28–§33, §38, §39) are developed in a companion paper [16].

7 Limitations

(i) The Jacobian singularity at $\theta \in \{0, \pi\}$; regularization costs exactness there. (ii) The arccos range: qg-space updates cannot reach $\theta < 0$. (iii) Z-basis blindness to phase-encoded entanglement and to readout dephasing. (iv) C_Z is not an entanglement measure. (v) qgs is not monotonic under T1. The pair $(qgs, q\bar{q}z)$ fixes this empirically, in 23 of 24 circuits, not provably. (vi) Finite-shot bias of qgs . (vii) Pole damping costs iterations and has a trapping mode. (viii) All device results, here and in the follow-up studies, are calibration-based or vendor-model simulations. Runs on real hardware (`-mode ibm`, IonQ) are prepared but have not yet been performed.

8 Code availability

The library, every example, and the regression tests pinning every number above (859 tests at the time of this revision) are available at <https://github.com/Viny2030/qang> (MIT license). Figure 1 is regenerated by `manuscript/make_figures.py`. Derivations with additional detail are in `RESEARCH_NOTES.md`.

Acknowledgments and use of AI tools

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A Riesz–Fréchet and why qg_Z is the canonical linear readout

Operators on $\mathbb{C}^{2^n}$ form a Hilbert space under $\langle A, B \rangle = \text{Tr}(A^\dagger B)$. By the Riesz–Fréchet representation theorem [21, 22], every linear functional on it has the form $\rho \mapsto \text{Tr}(O\rho)$ for a unique O . In finite dimension this is elementary linear algebra, so the appendix records a foundation rather than a new result. The functional $\rho \mapsto qg_Z(\rho)$ is represented by σ_z , and $q\bar{g}_Z$ by $\frac{1}{n} \sum_i Z_i$. Every linear readout, including linear XEB, has this form and hence an unbiased sample-mean estimator. qg_S is strictly concave in ρ , so no operator represents it, and its plug-in estimator is biased by Jensen’s inequality. By Proposition 1, qg_S is a fixed scalar function of the linear quantity qg_Z , so it carries no information about a single qubit beyond what σ_z extracts. (Riesz’s 1918 lemma on almost-orthogonal vectors concerns infinite-dimensional spaces and plays no role here.)

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Table 3: Follow-up studies (section numbers of `RESEARCH_NOTES.md`).

§	topic	main number	gain
15	natural gradient, cut cost, few-shot qg_Z	QFI = $1/(1 - qg_Z^2)$; cut $\gamma = 1 + 2\sqrt{1 - qg_Z^2}$; Haar prior 92% vs 8% coverage near a pole	closed forms; vs delta method
16, 19	QML encoding $\theta = \arccos x$	Chebyshev models fit polynomials to 10^{-13} vs 5×10^{-5} ; $\sim 10\times$ with shots	vs angle encoding [17]; classical Chebyshev fit wins
17	control grid uniform in qg_Z	probability error $\pi/2$ smaller; distribution loading TVD lower in 43/45 cases	vs θ grid; loses on fidelity near poles
18	T1 vs unital noise from counts	accuracy 0.88–0.91 vs 0.70–0.76 (XEB, HOP), strong noise only	vs XEB/HOP
20	IonQ noise models	H ₂ + qg_Z filter ~ 12 vs ~ 30 mHa, 2–3% shots dropped	vs readout mitigation
21	H ₂ electron-number witness	5.5 vs 18 mHa (HF 20.3); $11\times$ below HF at 2.5 Å; no gain on 6-qubit LiH	vs readout mitigation, HF
22	ODE solver [18]	exponential decay up to $17\times$ better; not in general	vs angle encoding; spectral method wins
23	local cost $(1 - q\bar{g}_Z)/2$	gradient variance $5600\times$ larger at 12 qubits; light cone trains 100 qubits classically	vs global cost [15]; classically simulable [19]
24	qg_Z filter vs ZNE by noise type	witness predicts when the filter works; ZNE + filter 2.0 mHa on fake_brisbane	complementary to ZNE
25	thermal qubit, $qg_Z = \tanh \beta h$	optimal thermometer $qg_Z \operatorname{artanh} qg_Z = 1$ [20]	none (prior, not qg)
26	which gates to cut	exact optimum over 3795 partitions; $2\times$ fewer shots than counting	vs angle-blind rules; same as find_cuts
27	repetition code choice under T1	$qg_Z + qg_X$ policy regret 2.7% vs 46–258%	decision rule
28	spin-resolved filters	23.2 $\rightarrow$ 14.7 mHa under depolarizing noise; none on fake_brisbane	conditional
29	1D Hubbard, 8 qubits	ZNE + spin filters 6–9 $\times$ below raw; wall at ≥ 188 ECR	vs raw, ZNE
30	QAOA, exactly K	filter doubles $P(\text{opt})$ of XY-QAOA; penalty QAOA $\approx$ random guess	vs QAOA; greedy wins
31	qubit characterization	unbiased T_{eff} (± 1 mK) and readout error, where the standard suite reads 0 K	vs standard calibration
32	Leung code vs T1	infidelity $\approx 0.92\gamma^2$ vs $\gamma/3$; rule: Leung iff $T_2 > T_1$	decision rule; code known
33	syndromes as witnesses	$\langle XXXX \rangle = qg_X^4$; drift tracking 13% regret with no probe shots	vs periodic recalibration
34	entropy gap = coherence	qg sum bound captures 92–100% of Z coherence (thermal Ising)	none on graph states
35	few-shot tomography	per-axis Haar prior best closed form on mixed states	loses on pure states; prior, not qg
36	Ramsey operating point	$F = (V^2 - qg_Z^2)/(1 - qg_Z^2)$	none (known rule)
37	Grover as $T_{2k+1}(qg_Z)$	$F_m = m^2/(1 - qg_Z^2)$; MLAE $\sim 9\times$ below Monte Carlo	none (known); noise must be modelled
38	IonQ noise models	filter doubles XY-QAOA; spin filters pay at depth	vs QAOA; greedy wins
39	native-gate ZNE on IonQ	CX folding is undone; at 2000 shots filter beats ZNE in RMS	depends on shot budget